

MACHINE LEARNING BASED MODEL FOR SOLAR RADIATION PREDICTION**Mohammed Eltahir Abdelhag¹, Salih Eltahir Elmarud Ali², Sherif Tawfik Amin³, Asif Ali Khan⁴, AbdElnasser AbdElwakil Metwally¹**¹Department of Information Technology and Security, Jazan University, Jazan 45142, Saudi Arabia²Department of Industrial Engineering, Jazan University, 45142 Jazan, Saudi Arabia³Department of Computer Science, Jazan University, Jazan 45142, Saudi Arabia⁴Department of Computer and Network Engineering, Jazan University, Jazan 45142, Saudi Arabiamohedhtahir@gmail.com¹, selmarud@jazanu.edu.sa², samin@jazanu.edu.sa³,asKHAN@jazanu.edu.sa⁴,ametwally@jazanu.edu.sa¹

In the era of an energy crisis, reliable and robust Solar Radiation (SR) prediction has become an integral part of thermal systems for renewable and clean energy production. Machine learning models are widely used as a precise effective solution in SR forecasting. In this paper, Artificial Neural Network (ANN) is employed to predict hourly SR. An open-source dataset available from the NASA hackathon task is used for evaluations. The models' effectiveness is evaluated using several evaluation metrics, and its efficiency is compared with Logistic Regression (LR) and AdaBoost models. The outcomes indicate that ANN model has a higher ability for predicting SR than LR and AdaBoost in the training and test stages.

Keywords: Solar radiation, machine learning, prediction, renewable energy, ANN

1. INTRODUCTION

The rapid developments in industrial has dramatically raised energy demand of our economy recently, and on the other hands energy resources are limited. The continually diminishing conventional resources like fossil fuels, petroleum, coal, and natural gas sources have further increased the demand-supply gap, known as the energy crisis. In search of green and sustainable energy, efforts are encouraged to use renewable resources like solar, wind, water, biomass, and geothermal. These renewable resources assist in not only reducing environmental pollution but also reducing stress on conventional resources. Solar is an easily and abundantly available renewable resource in most parts of the world. In recent years, Solar Radiation (SR) prediction has gained special attention in academia and a wide range of ML models have been used to forecast SR [1–5]. In [6], authors reported Support Vector Machine (SVM) and Extreme Gradient Boosting (XGBoost) to estimate SR using ML models. Comparative evaluation using prediction accuracy, stability, and computational time showed that XGBoost outperformed SVM for estimating SR. In [7], the authors assessed the ability of kernel-based nonlinear extension of Arps decline and Gradient Boosting (GB) models with categorical features support to predict daily global SR. It was reported that both models are suitable to be used in SR prediction. In another work [8], bagging and boosting ensemble models improve the performance of Support Vector Regression (SVR), decision trees, and ANN models for hourly SR prediction. The results demonstrated that ensemble models are capable of improving SR

prediction compared to other models. ANN has been widely used in predicting SR using different metrological and geographical [9,10]. In [11], authors investigated ANN and Multi Linear Regression (MLR) models for SR prediction from two stations using temperature, humidity, and wind velocity. Results showed that both models were consistent. In [12], ANN, SVM, deep learning, and Kernel Nearest Neighbor (KNN) are compared for daily SR prediction for the Turkish state meteorological service. It was reported that the ANN model outperformed SVM, deep learning, and KNN. In [13], an ANN model based on the Levenberg–Marquardt feed-forward and the back-propagation algorithm was used for the SR prediction of four metro cities in India. In [14], ANN, SVM, and genetic programming are compared for SR prediction using several stations in Australia. The results showed that ANN achieved better prediction outcomes than those of other models. In [15], ANN and Multi Linear Regression (MLR) are compared for SR. Results indicated that both ANN and MLR produced accurate estimation results for SR. In another work [16], ANN combined with a Genetic Algorithm (GA) for predicting daily SR. the authors used the duration of sunshine hours and air temperature as input to the models. The results showed that ANN combined with GA achieved better results than ANN alone. In [17], the performance of ANN, Multivariate Adaptive Regression Spline (MARS), Adaptive Neuro-Fuzzy Interface System (ANFIS), Regression Tree (CART), and Gradient Boosting Tree (GBT) is investigated for SR prediction. The comparative results showed that GBT performed better than the other models. The main goal of this paper is to test the suitability of the ANN model for forecasting SR using a publically available SR dataset. The performance of the ANN is evaluated using Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), coefficient of determination (R^2 measurements during training and testing phases and is compared with LR and AdaBoost models. The remaining parts of this paper is organized as follows: Section 2 gives an overview of both dataset and ANN model. The statistical measurements which are used to evaluate the ML models are presented in section 3. Results and analysis of the ANN and other models is provided in section 4, and finally section 5 concludes the paper.

2. MATERIAL AND METHOD

2.1. Study case

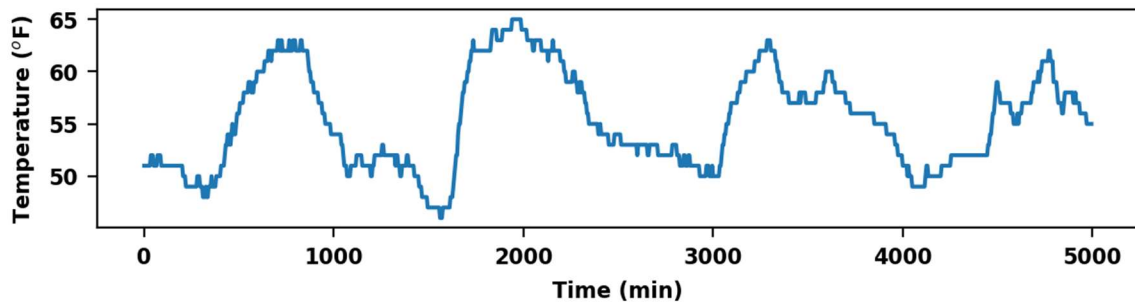
A dataset from the HI-SEAS weather station is used for evaluation in this work. It is freely available as a part of the NASA hackathon task and comprises eleven attributes as shown in Table 1. The UNIX time is useful in sorting the recorded measurements.

Table 1. Description of the SR dataset

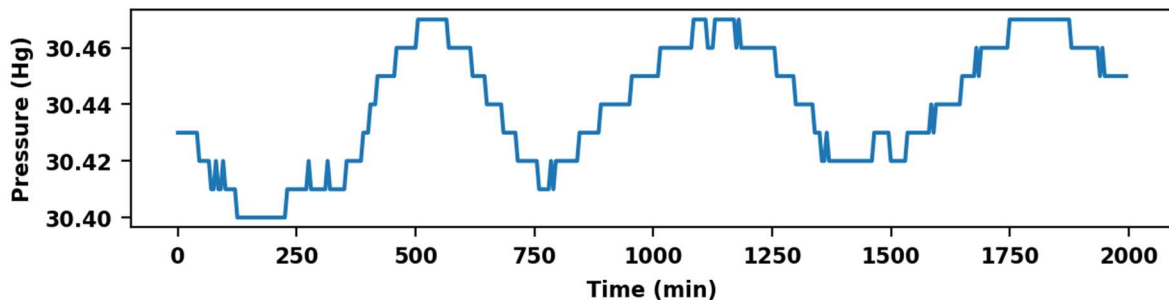
Attribute	Format
UNIX time of recording	seconds since Jan 1, 1970
Date of recording	yyyy-mm-dd
Local time of recording	hh:mm:ss 24-hour
Temperature	Fahrenheit

Humidity	Percent
Barometric pressure	Hg
Wind direction	Degrees
Wind speed	Miles per hour
Sunrise time	Hawaii time
Sunset time	Hawaii time
Solar radiation	Watts per sq. meter

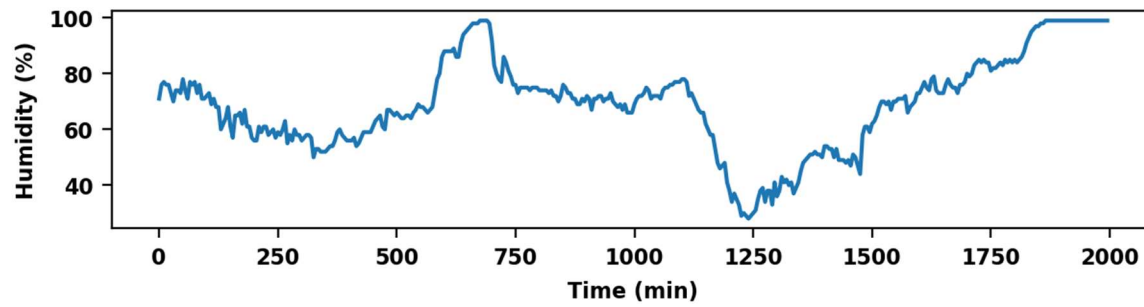
UNIX time of recorded measurement is processed to generate two columns corresponding to the month and year of the recording. The local recording time was processed to generate the hour of the day as an attribute. Finally, a difference between sunrise and sunset times is calculated as the total time of the day during which the sun is available. SR values are converted to kiloWatts per square meter to reduce large variations that may decrease model performance. An example of various features as a function of time is shown in Figure 1. The independent axis is time, and the data is shown for the first 2000 minutes (33 hours and 20 minutes). Fig. 1(a)–(c) shows low-frequency variation in temperature, pressure, and humidity with evident daily periodic nature for the first two. Wind speed and direction show no periodic nature.



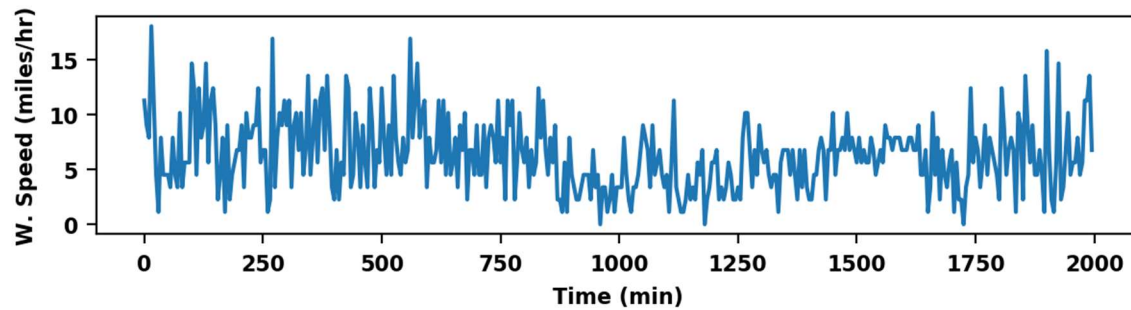
(a)



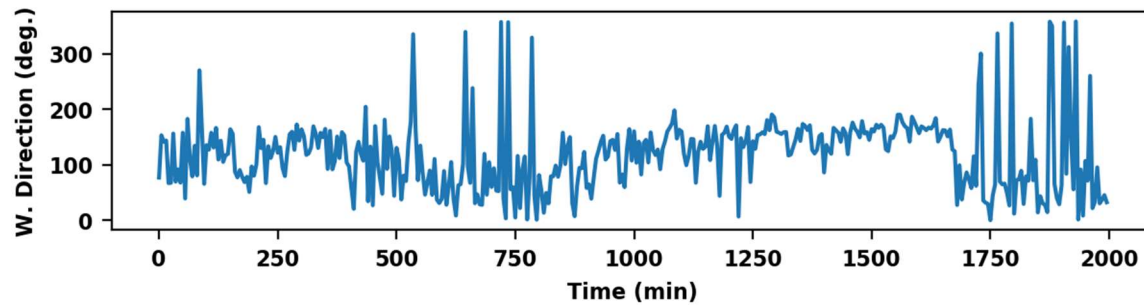
(b)



(c)



(d)



(e)

Figure1. Time variability of (a) temperature, (b) pressure, (c) humidity, (d) wind speed, and (e) wind direction.

2.2. ANN

The ANN model is initially introduced to mimic human brain, it is an excellent model for learning the non-linear relation among input and output features [18]. The Multilayer Perceptron (MLP) architecture of ANN consists of: An input layer, an output layer and one or more hidden layers, which can contain multiple neurons activated by a non-linear function for the purpose of transforming a weighted combinations of the output from the previous layer.

Based on the dataset and its nature, the input and output layers' neurons. Are specified In the current work, nine input nodes according to the used dataset attributes and one output to forecast SR are

used. All the attributes are normalized and standardize individually to have (i.e, zero mean and unit variance). Then they are recorded to apply an inverse transformation on the estimated SR values, two- hidden layers (10, 5 neurons) and rectified linear-unit is employed as an activation function and finally we trained the model using back-propagation with Adam optimizer to reduce forecasting error.

3. EVALUATION MEASUREMENTS

The following measurements are used to measure the performance of the ANN and other models. These measures are widely used in several works for SR prediction [14, 17, 19, 20].

$$MAE = \frac{1}{N} \sum_{i=1}^N |m_i - p_i|, \quad (1)$$

$$MSE = \frac{1}{N} \sum_{i=1}^N (m_i - p_i)^2, \quad (2)$$

$$RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^N (m_i - p_i)^2}, \quad (3)$$

$$R^2 = 1 - \frac{\sum_i (m_i - p_i)^2}{\sum_i (m_i - \mu_m)^2}, \quad (4)$$

where, m_i is the i th measured SR data, N is the number of observations, p_i is the i th SR predicted data by the models and μ_m is the mean value of the measured SR.

4. SIMULATION RESULTS AND DISCUSSION

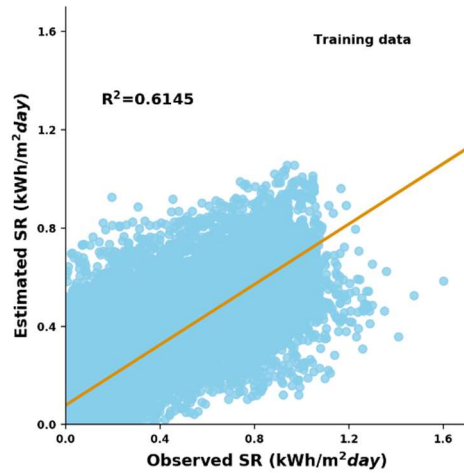
The ANN, and the other comparative models are assessed using the statistical measures. The results on the training and testing phases are given in Table 2 and provided in this section.

Table 2. Comparative results of SR prediction during training and testing by each model

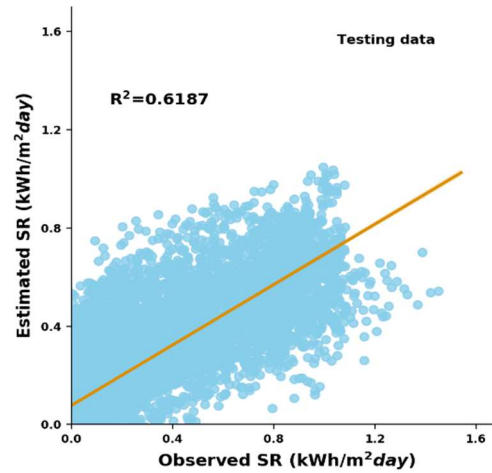
Model	Dataset	MAE	MSE	RMSE	R ²
LR	Training	0.1476	0.0383	0.1956	0.6145
	Testing	0.1484	0.0385	0.1962	0.6187
AdaBoos t	Training	0.0638	0.0130	0.1041	0.8772
	Testing	0.0645	0.0133	0.1154	0.8543
ANN	Training	0.0259	0.0036	0.0607	0.9678
	Testing	0.0425	0.0094	0.0974	0.9129

From the results provided in Table 2, the ANN gained the lowest results than those of LR and Ad boost in terms of MAE, MSE, RMSE and the highest value of R², which indicate that ANN is an

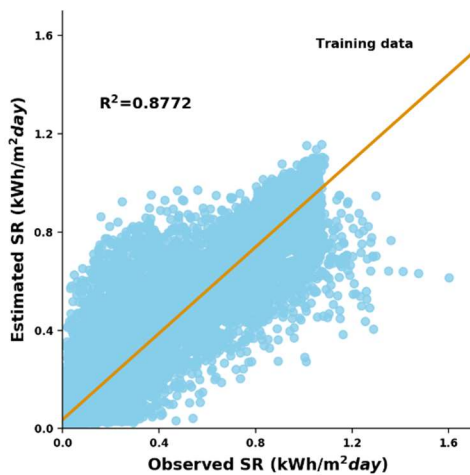
appropriate model for forecasting SR than LR and AdaBoost. Figure 3 illustrates relation between predicted SR values using the ANN, and the other comparative methods in training and testing phases. It can be seen that the ANN model has the highest correlation of 0.9678 and 0.9129 in the training and testing respectively and the daily SR data by the ANN is closer to the observed ones during the phases of the tainting and testing, while the AdaBoost showed better correlation than the LR model. From both the performance measurements and scatterplots, it is evident that ANN has great power to forecast SR much better than the other competitor approaches.



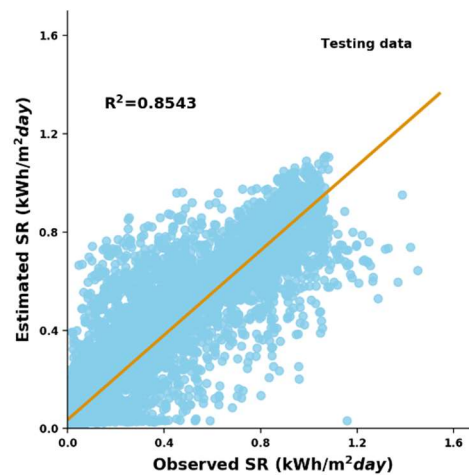
(a)



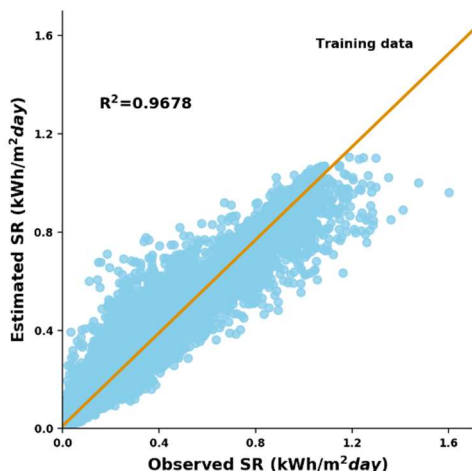
(b)



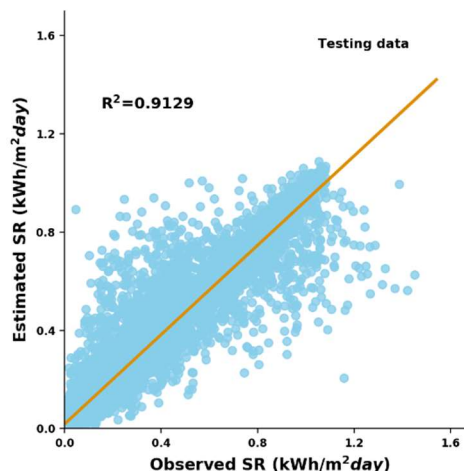
(c)



(d)



(e)



(f)

Figure 3. Scatterplots of SR data by the LR ((a)-(b)), AdaBoost ((c)-(d)), and ANN ((e)-(f))

5. CONCLUSION AND FUTURE WORK

SR prediction necessitates ML techniques to be utilized due to their capability to predict reliable and robust SR values. The work investigated the capability of ANN model: for forecasting SR and assess its performance using five widely used statistical measurements. The results indicated that ANN model obtained reliable and accurate SR prediction compared to LR and AdaBoost models: In the future, the plan is to work and use metaheuristic methods to pick the most informative features in the application of SR due to the great potential of these methods in other domains.

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